Cloud FinOps & Cost Optimization Strategies

Kubernetes Right-Sizing, Spot Instances, and Financial Accountability

Abstract

Cloud elasticity frequently leads to uncontrolled expenditure without rigorous financial governance. This paper examines FinOps principles, Kubernetes cost allocation, spot instance management, and unit economic optimization.

Keywords: FinOps, Cost Optimization, Kubecost, Spot Instances, Unit Economics.

Table of Contents

- 1. Introduction to Cloud FinOps and Financial Accountability
- 2. Kubernetes Right-Sizing: Requests, Limits, and VPA Optimization
- 3. Leveraging Spot Instances and Preemptible VMs safely
- 4. Namespace Cost Allocation and Team-Level Accountability
- 5. Egress Cost Minimization and Network Architecture Optimization
- 6. Storage Lifecycle Management and Unattached Volume Cleanup
- 7. Automated Scheduling and Off-Hours Scale-Down Policies
- 8. Multi-Cloud Cost Arbitrage and Pricing Comparison Models
- 9. Kubecost, OpenCost, and Cloud Provider Billing Integration
- 10. Unit Economics: Calculating Cost per Active User and Transaction
- 11. Budgeting, Anomaly Detection, and Automated Cost Alerting
- 12. Enterprise Governance: Tagging Policies and Resource Ownership
- 13. Case Studies: Reducing Cloud Spend by 40% Without Sacrificing SLA
- 14. Integrating FinOps into Engineering Pull Request Workflows
- 15. Future Outlook: AI-Driven Autonomous Cost Optimization